

Distinct Values Subarrays

Time limit: 1.00 s

Memory limit: 512 MB

Given an array of n integers, count the number of subarrays where each element is distinct.

Input

The first line has an integer n : the array size.

The second line has n integers $x_1, x_2, \dots, x_n$: the array contents.

Output

Print the number of subarrays with distinct elements.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
4 1 2 1 3	8

Explanation: The subarrays are $[1]$ (two times), $[2]$, $[3]$, $[1, 2]$, $[1, 3]$, $[2, 1]$ and $[2, 1, 3]$.